

Unit

2

Lesson

2



Innovators in Action.

Smart Moves.

Time Frame: [90 min].

Ready, Set, Learn!

You Need to Know

- How an IR sensor works for motion detection.
- Integrate the IR sensor and servo motor to automate the bin lid.
- Make the servo motor move when an object is detected.

Challenge questions of the lesson.

1. How can you design the automatic lid of your smart trash bin (which uses the IR sensor to detect and the servo motor to open) so that it's easy and safe for everyone to use? Think about people of different heights (like young children or adults) or those who might be using a wheelchair. What features or adjustments could you add to ensure inclusive access?
2. If the smart trash bin's lid opens and closes by itself with a motor, how can you make sure it operates at a safe and appropriate speed – not too fast to surprise someone, but not too slow either? What kind of controls or additional sensor inputs might be needed to make the automatic lid even safer and more user-friendly?

SEL Relation

- **Inclusive Design (Empathy & Social Awareness):** Students will practice considering the needs and perspectives of diverse users, including individuals with varying heights or those using wheelchairs, in their design process for the smart trash bin's automatic lid.
- **Collaboration and Communication (Relationship Skills):** Students will share their design ideas for the smart bin, listen to the ideas of others, and work together (if in groups) to develop their concepts for the automatic lid and sorting mechanism.
- **Creative Problem-Solving (Self-Awareness & Responsible Decision-Making):** Students will need to think creatively to solve the challenges of accessibility, reliable sensor detection, and smooth motorized lid movement for the smart bin, making responsible decisions about their design choices.
- **Perspective Taking (Social Awareness):** Students will be encouraged to think about how different individuals would experience their smart trash bin's automatic lid design and how effectively it meets their needs.

Engage



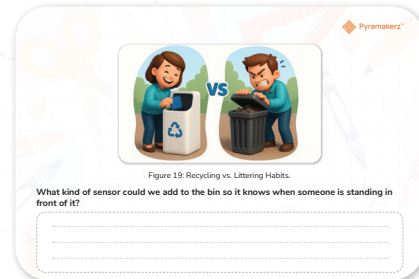
Teacher Role:

- **Storyteller & Connector:** Guide students through the dialogue with Mrs. Sarah, Laila, Adam, and Student(s) to spark curiosity about automating the smart bin and connect it to how technology solves real-world problems like keeping things clean.
- **Facilitator of Wonder:** Ask open-ended questions like:
 3. "How does the IR sensor act like the 'eyes' of our bin?"
 4. "Why is having an automatic lid a good idea for a trash bin?"
- **Clarifier:** Explain the roles of the key components mentioned (Arduino, servo motor, IR sensor) in simple terms, suited for early learners, emphasizing how electricity powers them.
- **Discussion Guide:** Lead students through activities or questions that prompt them to think deeply about automation, sensors, and the benefits of their smart bin idea:
 - "Which of these parts (Arduino, servo, IR sensor) is like the 'brain,' the 'eyes,' and the 'muscle' of our bin?"
 - "How does having an automatic lid encourage everyone to use the bin properly?"

Expected From Students:

- **Active Listening & Dialogue Engagement:** Students will listen carefully to the dialogue, focusing on the roles of Arduino, servo motors, and IR sensors in the smart bin idea.
- **Prediction & Problem-Solving:** They will begin asking questions like:
 1. "How does electricity get from the Arduino to the servo motor?"
 2. "What other sensors could we use?"
- **Concept Justification:** Students will discuss and explain why each chosen component (Arduino, servo motor, IR sensor) is important for making the lid automatic and keeping the bin clean.
- **Innovation & Design Thinking:** They will consider how technology can improve everyday objects like trash bins, laying the foundation for their own design choices.

Parts of the books included.



Story



1. Set the Scene with Energy and Curiosity.

- Teacher says (with a tone of energy and curiosity): "Today, we're diving even deeper into an awesome idea: making our smart trash bin open its lid all by itself! No more sticky fingers! Remember how we figured out electricity can make things move? Well, Adam and Laila, along with their classmates, have been brainstorming how to apply this to our bin's lid. Let's listen in on their latest discussion!"
- Dialogue Reenactment (or Teacher reads with roles): Mrs. Sara: Alright, let's talk more about your idea for the automatic bin lid. Laila: We already have a plan! Last time, we learned how electricity can make the bin work by itself—and we even talked about Arduino, that tiny computer. Zayn: And the servo motor! It can move the lid up and down. Adam: Yes! But to make it open by itself, we need something that tells it when someone is there!
- Pause and ask: "Have you ever seen an automatic door open when you walk close? What do you think makes it do that? What do you think Adam is talking about when he says 'something that tells it when someone is there?'" (Guide students to the concept of a sensor, specifically an IR sensor if they don't get there directly). This gets students thinking about sensors and automation in their daily lives.
- Ask Guiding Questions: Prompt students, now that the components (IR sensor, Arduino, servo motor) have been introduced:
 - "How does each part — the IR sensor, the Arduino (the tiny computer), and the servo motor (the one that moves the lid) — play a special role in making the lid open automatically?"
 - "Why is an automatic lid a good solution for a public trash bin?"
 - "What happens if one of these parts isn't working?" These questions lead naturally into deeper analysis of the system.

- Transition to Activity: Say: "Fantastic thinking! Now that we know about the key parts – the sensor, the Arduino 'brain', and the servo 'muscle' – that will make the lid automatic, let's think like designers and problem-solvers about how we'll build this and make sure all our electronic parts work together!"

Possible Strategies for Implementation:

1. System "Flow" Charting (or Drawing the Story):

- How to Implement: After reading the dialogue and clarifying the roles of the components, give students a large piece of paper or a whiteboard space. As a class or in small groups, have them draw a simple "flow" chart or picture story of how the smart bin lid opens. They should draw the IR Sensor first (detecting someone), then an arrow to the Arduino (the brain getting the signal), then another arrow to the Servo Motor (the muscle moving the lid).
- Why it Helps: This visual mapping helps students understand the sequential relationship between components and how they work together as a complete automated system, moving from abstract ideas to concrete connections.

2. Action Vocabulary & "Show Me!":

- How to Implement: Introduce and discuss new words: IR sensor, Arduino, servo motor, signal, automatic. For each word, don't just add it to a word wall – have students physically demonstrate what it does or draw a quick symbol for it. For example, for "IR sensor," they might put their hand out to "detect" something; for "servo motor," they might move their arm to an exact angle; for "Arduino," they might point to their head as the "brain."
- Why it Helps: This multi-sensory approach deepens understanding of technical vocabulary, supports kinaesthetic learners, and makes the learning stick by linking words to actions and visuals.

Let's Think!



Objective:

Students will analyze the benefits of using hands-free trash bins and explore how various technologies (like smart cans, robots, sensors, and apps) contribute to keeping public spaces cleaner and more comfortable. They will develop critical thinking and communication skills by explaining the specific roles of these smart components and advocating for user-friendly designs.

Answers:

1. **What makes using a hands-free trash bin more comfortable or enjoyable for people? Why might this type of bin be used more often than regular ones?**
 - It's more hygienic because you don't have to touch a dirty lid.
 - It's convenient when your hands are full.
 - People might use it more often because it's easier and cleaner, so trash ends up in the bin instead of on the ground.

- It feels modern and cool to use.
- 2. What kind of sensor could we add to the bin so it knows when someone is standing in front of it?**
- An IR (Infrared) sensor could be added to the bin. It sends out a beam of light, and when something (like a person's hand or body) gets close, the light bounces back to the sensor, telling the bin that someone is there.

Possible Strategies for implementation:

1. "Bin Bother" vs. "Bin Bliss" Think-Pair-Share:

- How to Implement: Start by asking students: "Imagine you're trying to throw something away, but the bin is overflowing, or it's sticky, or you don't want to touch it. How does that make you feel?" (Give think time). "Now, imagine a bin that opens magically for you. How does that feel?" (Pair to share thoughts). "Share with the class why one experience is better than the other, and what makes a hands-free bin so appealing!"
- Benefit: Builds empathy for users and helps students articulate the practical comfort and hygiene benefits of hands-free technology from a personal perspective, directly addressing the first question.

2. Sensor Spotlight & Brainstorm:

- How to Implement: This strategy directly supports students in answering Question 2. Introduce the concept of sensors that detect presence. Ask students directly: "What kind of sensor could we add to the bin so it knows when someone is standing in front of it?" Encourage them to brainstorm different types of sensors they might know (e.g., motion, infrared, ultrasonic). Discuss how each type works briefly and which would be most suitable for detecting a person approaching the bin.
- Benefit: Focuses students' thinking on specific technological solutions, enabling them to identify a suitable sensor and understand its function for automated features, directly addressing the second question.

3. "Why Go Smart?" Persuasive Argument:

- How to Implement: After students have explored the comfort/usability aspects and identified a sensor, challenge them to synthesize their understanding. Ask: "Based on what we've discussed about hands-free bins and sensors, how would you convince someone that a smart, hands-free trash bin is a necessary improvement for our school or public spaces?" Encourage them to use vocabulary from their discussions about comfort, hygiene, and sensor functionality.
- Benefit: Encourages students to synthesize their understanding of both user benefits and technological solutions, practicing persuasive communication skills by advocating for their ideas based on the two task questions.

Explorer's Toolkit



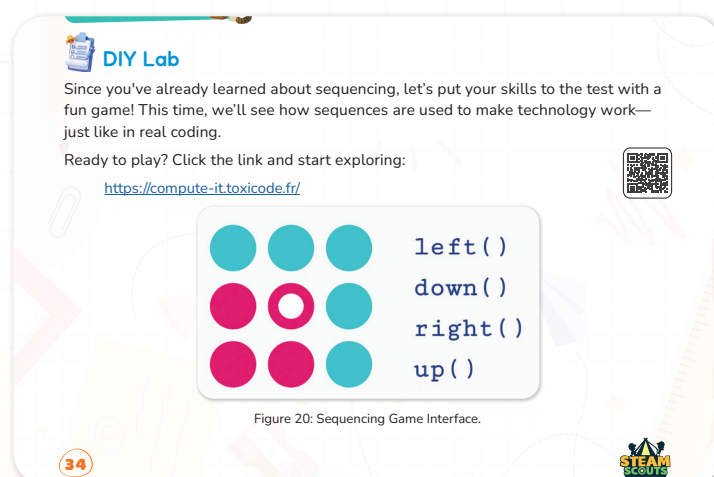
Teacher Role:

- **Game Introducer & Sequence Explainer:** Get students excited about the game! Introduce the core concept of a sequence as "doing things in order" and explain why order is super important in coding – it's how we tell computers exactly what to do, step-by-step.
- **Game Navigator & Guide:** Help students get started with the "Compute It!" game. Guide them through the interface, explaining how to drag and drop commands, run their code, and understand the feedback.
- **Puzzle Troubleshooter:** Be ready to assist students if they get stuck on a particular puzzle. Instead of giving answers, prompt them with questions like: "What happens if you swap these two steps?" or "What's the very first thing the character needs to do?"
- **Concept Questioner:** Encourage deeper thinking as students play. Ask: "Why did your character move that way when you put those blocks in that order?" "What would happen if you put the 'turn' command before the 'move' command?" This helps them connect the game to real coding logic.

Expected From Students:

- **Active Gaming & Problem-Solving:** Students will actively engage with the "Compute It!" game, trying out different sequences to solve the puzzles.
- **Following Game Logic:** Students will follow the game's instructions and challenges to correctly arrange commands and achieve the desired outcomes.
- **Observing & Debugging:** Students will carefully observe what happens when they run their code sequences, identify if something went wrong (a "bug"), and then try to fix it by changing the order of commands.
- **Understanding Sequence Power:** Students will begin to understand that the precise order of commands (the sequence) is crucial in coding for making technology work exactly as intended.

Parts of the books included.





DIY Lab

Objective:

Students will understand and apply the concept of sequences in coding by playing an interactive online game. Through this engaging challenge, they will learn that commands must be given in a specific order to make technology (like our smart bin or any robot!) work exactly as intended. This activity builds foundational coding logic and problem-solving skills.

Materials Needed:

- Laptop for each student or pair.
- Internet Access
- Link to the Game: <https://compute-it.toxicode.fr/>



Activity Instructions:

1. Game Launch & Mission Intro: "Alright, coders! Get ready to unlock the secrets behind how all your tech works! Today, we're jumping into a super cool game where we'll learn about sequences. Guess what? Sequences are super important in coding! It means doing things in the right order, just like following a recipe."
2. Guide students to open the game link on their devices.
3. Meet Your Character & Commands: "In this game, you'll control a little character. Your mission is to get them to the goal by giving them commands. Look at the command blocks available. What do they make your character do? ('move forward,' 'turn left,' 'turn right')."
4. Show how to move the point using the keyboard arrows.
5. Explain: "That's your first sequence! You told the computer exactly what to do, step by step."
6. The Power of Order (Why Sequence Matters!): "What if we need to turn? If you want to move forward and then turn right, what order should the commands be in? What happens if you put 'turn right' before 'move forward'?"
7. Encourage them to try different orders to see the impact. "The computer does exactly what you tell it, in the order you tell it!"
8. Tackling the Puzzles: "Now you're ready for the puzzles! Each level has a specific goal. Your job is to create the perfect sequence of commands to reach that goal. Think carefully about each step!"

Possible Strategies for Implementation:

1. "Read the Goal First!" (Mission Briefing):
 - How to Implement: Before students make any move any, have them explicitly identify what the character needs to do on that specific game level. "What's the finish line? What objects do you need to collect?"
 - Why it Helps: This ensures students understand the problem they are trying to solve, making their coding efforts more purposeful and efficient.

2. "Step-by-Step Think Aloud!":

- **How to Implement:** Encourage students (especially for trickier puzzles) to verbally "narrate" the path their character should take, one action at a time, before making any move. "Okay, first I need to move one step. Then, I need to turn left. Then, I move another step..."
- **Why it Helps:** Breaking down complex tasks into smaller, sequential steps makes the problem more manageable and helps in logically ordering the commands.

3. "Test, Observe, Fix! (Be a Code Detective!)":

- **How to Implement:** Emphasize the iterative nature of coding. Encourage students to run their current sequence frequently, observe exactly what the character does (even if it's wrong), and then make small adjustments to their code based on those observations. Teach them to look for where their sequence "went wrong."
- **Why it Helps:** This promotes a hands-on, experimental approach to coding, where "mistakes" are valuable learning opportunities for debugging and refining their logic.

4. "Predict the Path!":

- **How to Implement:** For a given set of command blocks, challenge students to predict the character's exact path before they hit the "play" button. They can even draw the predicted path on paper.
- **Why it Helps:** This sharpens their ability to "read" code and mentally simulate its execution, strengthening their understanding of how sequence directly dictates outcome.

Explain



Teacher Role:

- **Concept Introducer:** Introduce the IR sensor as a key "sense" for our smart projects, comparing it to how our eyes see, but with invisible light (infrared). "Today, we'll learn how our smart bin's 'eyes' – the IR sensor – actually work!"
- **Video Narrator & Explainer:** Guide students' viewing, pausing the video to clarify concepts like the IR LED (transmitter) emitting light, the photodiode (receiver) catching reflections, and how light/dark surfaces reflect differently.
- **Scientific Explainer:** Explain the basic physics of light reflection in simple terms, connecting "brighter colors reflect more" and "darker colors absorb more" to how the sensor detects.
- **Application Connector:** Constantly link the sensor's function back to real-world examples, especially our smart trash bin ("How does this sensor help our bin know when someone is near?").
- **Discussion Moderator:** Lead a post-viewing discussion, encouraging students to share their observations, ask questions, and predict other uses for IR sensors.

Expected From Students:

- Active and Focused Viewing: Students will watch the video attentively, concentrating on the visible and invisible parts of the IR sensor and how it responds to surfaces.
- Detailed Observation: Students will carefully observe how the IR transmitter emits light and how the receiver catches reflections, noting the impact of different colored surfaces.
- Understanding IR Sensor Principles: Students will begin to grasp the core concepts of how an IR sensor detects presence or distance by using infrared light.
- Analytical Engagement: Students will think critically about the roles of the transmitter (sending light) and the receiver (catching light) in the sensor.
- Thoughtful Participation: Students will actively contribute to pre- and post-viewing discussions, sharing observations and ideas for IR sensor applications.
- Connection to Application: Students will connect the video's concepts to how the IR sensor will enable the automatic lid of our smart trash bin and other devices.

Parts of the books included.



Screen Lessons

Click on the link to learn more about IR sensor!

https://www.youtube.com/watch?v=hVej7tx-naw&ab_channel=CircuitClass

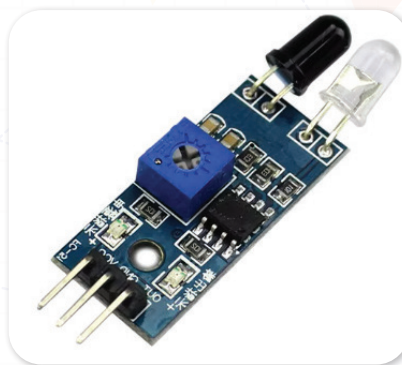


Figure 21: IR Sensor.



Screen Lessons

Objectives:

Students will understand how an IR (Infrared) sensor works, including its main components (IR transmitter and receiver) and how it detects objects based on reflected light and color. Through video analysis and discussion, they will grasp the concept of infrared technology and recognize its role as a "sensing" component in smart systems like our automatic trash bin.

Video link: https://www.youtube.com/watch?v=hVej7tx-naw&ab_channel=CircuitClass



Possible Strategies for implementation:

1. Pre-Watching: "What are 'Machine Eyes'?" Brainstorm!

- How to Implement: Before showing the video, ask students: "How do we know when something is close to us? (Our eyes, touch). How do machines know? What are some 'machine eyes' or 'machine ears' you've seen?" Write down their ideas. Introduce the IR sensor as a type of "machine eye" that uses invisible light.
- Why it Helps: Activates prior knowledge about senses, builds an analogy for understanding sensors, and sets a clear purpose for watching the video about the IR sensor.

2. During Watching: "IR Sensor Detective Checklist!":

- How to Implement: Provide students with a simple checklist to complete as they watch the video. Items could include:
 - "What are the two main parts of an IR sensor?" (Transmitter, Receiver)
 - "What kind of light does it use?" (Invisible/Infrared)
 - "What happens when IR light hits a bright surface?" (Reflects more)
 - "What happens when IR light hits a dark surface?" (Absorbs more)
 - "What are two places you might see an IR sensor used?"
- Why it Helps: Encourages active viewing, helps students focus on key details about the sensor's components and working principle, and aids in information retention.

3. Post-Watching: Discussion and Reflection.

- Strategy 1: "Draw the Invisible Light!":
 - How to Implement: After the video, have students draw a simple diagram of an IR sensor next to a surface. They should draw the invisible IR rays going out and reflecting back. They can draw a bright surface vs. a dark surface to show different reflections.
 - Benefit: Solidifies understanding of the sensor's physical components and its working principle (emission and reflection) through visual representation.
- Strategy 2: "Real-World Sensor Spotting!":
 - How to Implement: Discuss real-world examples where IR sensors are used (e.g., automatic doors, TV remotes, burglar alarms, line-follower robots, hand dryers). Ask: "How does the IR sensor help these things work?"
 - Benefit: Helps students connect the abstract concept of an IR sensor to tangible applications, making the learning relevant and transferable to their daily lives.
- Strategy 3: "Our Smart Bin's 'Eyes' Discussion":
 - How to Implement: Specifically focus the discussion back to the smart trash bin project. "How will the IR sensor help our smart bin know when to open its lid? What are the benefits of the bin having 'eyes'?"
 - Benefit: Reinforces the direct application of the learned concept to their main project, ensuring understanding of its specific role in their design.

Elaborate



Teacher Role:

- **Concept Reinforcer:** Briefly review the core concepts of the IR sensor, reminding students about its "invisible light" (infrared), its transmitter and receiver, and how it detects objects or movement. Connect it back to the "eyes" of our smart bin.
- **Question Facilitator:** Guide students through each multiple-choice question about the IR sensor. Read the question and all options aloud clearly.
- **Critical Thinking Prompter:** Encourage students to think deeply about why an answer is correct or incorrect. Ask: "Why do you think 'B' is the best answer for what an IR sensor does, based on what we learned?" or "What's wrong with option 'C'?"
- **Discussion Leader:** After students have considered their answers, facilitate a class discussion for each question. Allow students to share their chosen answer and, most importantly, explain their reasoning.
- **Concept Clarifier:** Address any misconceptions or confusion that arise during the discussion, ensuring a solid understanding of the IR sensor's purpose and how it communicates signals.

Expected From Students:

- **Analytical Readers:** Students will carefully read and analyze each question and all provided answer choices.
- **Knowledge Appliers:** Students will use their understanding of how IR sensors work (from the video and previous discussions) to select the best answer for each question.
- **Reasoning Articulators:** Students will be prepared to explain why they chose a particular answer, demonstrating their comprehension of the IR sensor's function and signaling.
- **Active Participants:** Students will contribute thoughtfully to the discussion, sharing their answers and listening to their classmates' reasoning, refining their own understanding.
- **Concept Confirmers:** By the end of the activity, students will clearly understand that an IR sensor detects proximity/movement using invisible light and communicates this detection by sending an electrical signal.

Parts of the books included.



Apprentice

1. An IR sensor is a key part of many automated machines. Which statement best describes what an IR sensor is designed to do?
 - A. It measures how warm or cold something is.
 - B. It can "see" when something moves or is nearby by using a type of light we can't see.
 - C. It creates the electricity that powers the system.
 - D. It sends direct commands to make a motor move.
2. When an IR sensor "sees" something in its path, what's its immediate way of telling the rest of the system, so something can happen?
 - A. It just stops working.
 - B. It sends a message (a signal) to turn on or activate the circuit.
 - C. It makes the power in the circuit weaker.
 - D. It completely restarts the entire system.



Figure 22: IR Sensor Detecting an Object.

STEAM SCOUTS

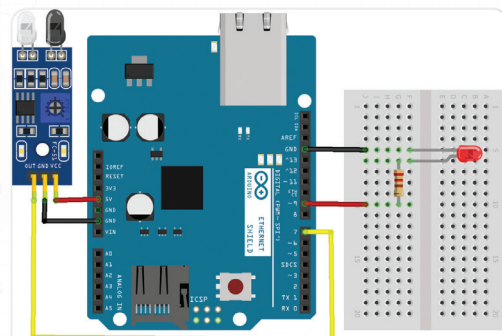
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Builder

Let's level up our circuit:

Create an automated light that turns on when the IR sensor detects an object or presence, and turns off when nothing is detected.



fritzing

Figure 23: Automated Light Circuit with IR Sensor and Arduino.



Apprentice

Objective:

Students will identify real-world examples of objects or systems that use IR (Infrared) sensors within their school environment (or daily life). They will demonstrate understanding of how an IR sensor detects presence or motion to enable a specific function in that object, through a labeled drawing and written explanation. This activity builds early understanding of applied sensing technology and smart systems.

Answers:



Apprentice

1. An IR sensor is a key part of many automated machines. Which statement best describes what an IR sensor is designed to do?
 - A. It measures how warm or cold something is.
 - B. It can "see" when something moves or is nearby by using a type of light we can't see.
 - C. It creates the electricity that powers the system.
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Figure 22: IR Sensor Detecting an Object.

STEAM SCOUTS

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Possible Strategies for Implementation:

1. "IR Sensor Scavenger Hunt Brainstorm!":

- How to Implement: Start by revisiting the key idea from the multiple-choice question: "An IR sensor 'sees' when something moves or is nearby using light we can't see." Then, ask students to brainstorm places or objects they've seen that seem to sense when something is near without being touched. Think about: automatic doors, hand dryers in bathrooms, TV remotes (even though those are transmitters, they're a good lead-in to IR light), paper towel dispensers. Write their ideas on a board.

- Benefit: Activates prior knowledge of automated devices and helps students connect the abstract concept of an IR sensor to tangible, real-world examples.

2. "What's it 'Seeing'? (The Sensor's Job)":

- How to Implement: For each object brainstormed (e.g., "automatic door"), ask guiding questions:
 - "What do you think the sensor on this object is 'looking' for?" (e.g., a person walking close)
 - "Why does it need to 'see' that?" (so, the door knows when to open for safety and convenience)
 - "How is this like what we learned an IR sensor does – seeing something nearby?"
- Benefit: Helps students analyze the specific function of the sensor within a larger system and link it directly to the definition of an IR sensor.

3. "Sensor Signal Story!":

- How to Implement: Once students choose one object that they believe uses an IR sensor, have them prepare for their drawing and explanation. Prompt them: "Draw your object. Then, draw the IR sensor on it. Now, show with an arrow how the IR sensor 'sees' something (like a hand). What 'message' or 'signal' does it send when it 'sees' something? And what does the object do then?"
- Benefit: Guides students in illustrating the input (detection by sensor), the signal (communication), and the output (action), demonstrating a deeper understanding of the sensor's role as asked in the multiple-choice questions.

4. "Explain Your Smart Spot!":

- How to Implement: After drawings are complete, have students present their chosen object. They should point to their drawing and explain: "This [object] has an IR sensor here. It detects [what it detects]. When it 'sees' that, it sends a [signal/message], and then the [object] [what it does]."
- Benefit: Encourages clear communication and consolidates their understanding by requiring them to verbalize the sensor's function and its impact on the object's behavior.



Builder

Objective:

Students will construct a working electronic circuit using an Arduino (the "brain" of the system), an IR sensor (the "eyes"), a servo motor (to "move parts"), an LED (as an indicator), and connecting wires (which "send electricity"). Through hands-on building and observation, they will explore how the IR sensor's input signals the Arduino, which then controls the servo motor's movement and the LED's state. This activity builds foundational skills in practical electronics, component integration, and understanding cause-and-effect within an automated system.

Materials Needed:

- Arduino Uno Board

- Breadboard
- IR Infrared Sensor Module
- Servo Motor
- LED (Light Emitting Diode)
- Resistor for LED (typically 220-330 ohm, not explicitly shown but good practice for LED)
- Jumper Wires (Assorted lengths, male-to-male and male-to-female as needed)
- (you can use tinkercad as well).

Activity Instructions:

Safety First! Remind students to handle all components carefully and ensure power is off (Arduino unplugged) before making connections.

1. Set Up the Power Hub (Breadboard & Arduino Power):

- Place the Arduino Uno board on your workspace.
- Place the breadboard next to it.
- Connect the 5V pin from the Arduino to the positive (+) power rail (usually red line) on the breadboard using a jumper wire.
- Connect a GND pin from the Arduino to the negative (-) ground rail (usually blue line) on the breadboard using a jumper wire.
- Remember: Wires "send electricity" through the circuit!

2. Connect the "Eyes" (IR Sensor):

- Carefully place the IR sensor module on the breadboard.
- Connect the VCC pin of the IR sensor to the positive (+) power rail on the breadboard.
- Connect the GND pin of the IR sensor to the negative (-) ground rail on the breadboard.
- Connect the OUT (signal) pin of the IR sensor to Digital Pin 7 on the Arduino.

3. Connect the "Muscle" (Servo Motor):

- Place the servo motor on your workspace. It usually has three wires: Power (often red/orange), Ground (often brown/dark), and Signal (often yellow/white).
- Connect the Power wire of the servo motor to the positive (+) power rail on the breadboard.
- Connect the Ground wire of the servo motor to the negative (-) ground rail on the breadboard.
- Connect the Signal wire of the servo motor to Digital Pin 9 on the Arduino.
- Remember: A servo motor is used "to move parts to a specific position"!

4. Connect the "Indicator Light" (LED):

- Insert the LED into the breadboard. Remember the longer leg is positive (+), shorter

is negative (-).

- Connect a resistor to the shorter (negative) leg of the LED.
- Connect the other end of the resistor to the negative (-) ground rail on the breadboard.
- Connect the longer (positive) leg of the LED to Digital Pin 13 on the Arduino. (Note: Pin 13 often has a built-in resistor, but for clarity and general good practice, indicating a resistor is beneficial, even if the diagram uses Pin 13 which often has one internally).

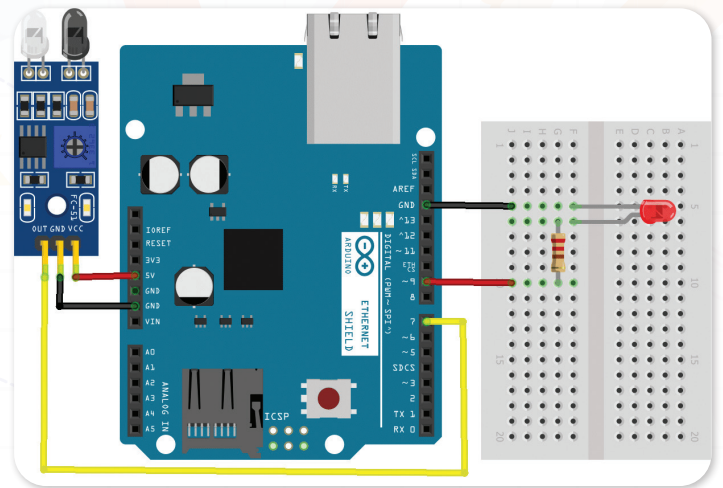


Figure 23: Automated Light Circuit with IR Sensor and Arduino.

Evaluate



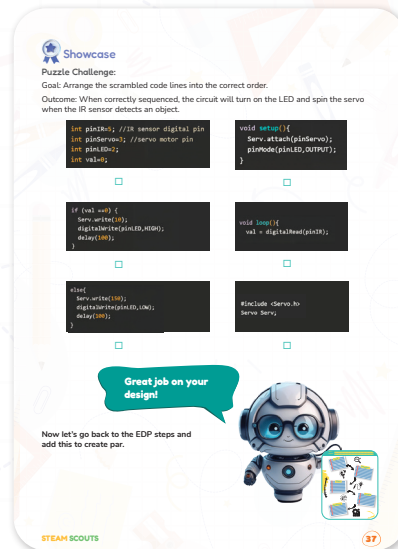
Teacher Role:

- Introduce the "Puzzle Challenge" as a logic and problem-solving activity, connecting it to how smart devices (like the automatic bin lid) use code to respond to their environment.
- Clearly explain the goal and the desired automated outcome of the code challenge.
- Guide students in analyzing the function of different code blocks (e.g., variable declarations, void setup(), void loop(), if/else statements) and how they relate to hardware components (IR sensor, LED, servo motor).
- Facilitate discussion and collaboration as students work to sequence the code correctly.
- Provide support and clarification as needed, helping students understand the flow of logic.

Expected From Students:

- Analyze individual scrambled code snippets to understand their purpose.
- Logically arrange the code lines into a correct sequence that achieves the specified automated behavior.
- Demonstrate an understanding of how an IR sensor's input can trigger actions from an LED and a servo motor through programming.
- Collaborate with peers to solve the coding puzzle.

Parts of the books included.



Showcase

Objective:

Students will apply logical reasoning to sequence scrambled code lines to create a functional program for an automated system.

Demonstrate understanding of how a sensor's input can control an LED and a servo motor through programmed logic.

Answer:

```

#include <Servo.h>

Servo Serv;

int pinIR = 5; // IR sensor digital pin
int pinServo = 3; // servo motor pin
int pinLED = 2; // LED pin
int val = 0;

void setup() {
  Serv.attach(pinServo);
  pinMode(pinLED, OUTPUT);
}

void loop() {
  val = digitalRead(pinIR);
  if (val == 0) {
    Serv.write(10);
    digitalWrite(pinLED, HIGH);
    delay(100);
  } else {
    Serv.write(150);
    digitalWrite(pinLED, LOW);
    delay(100);
  }
}

```

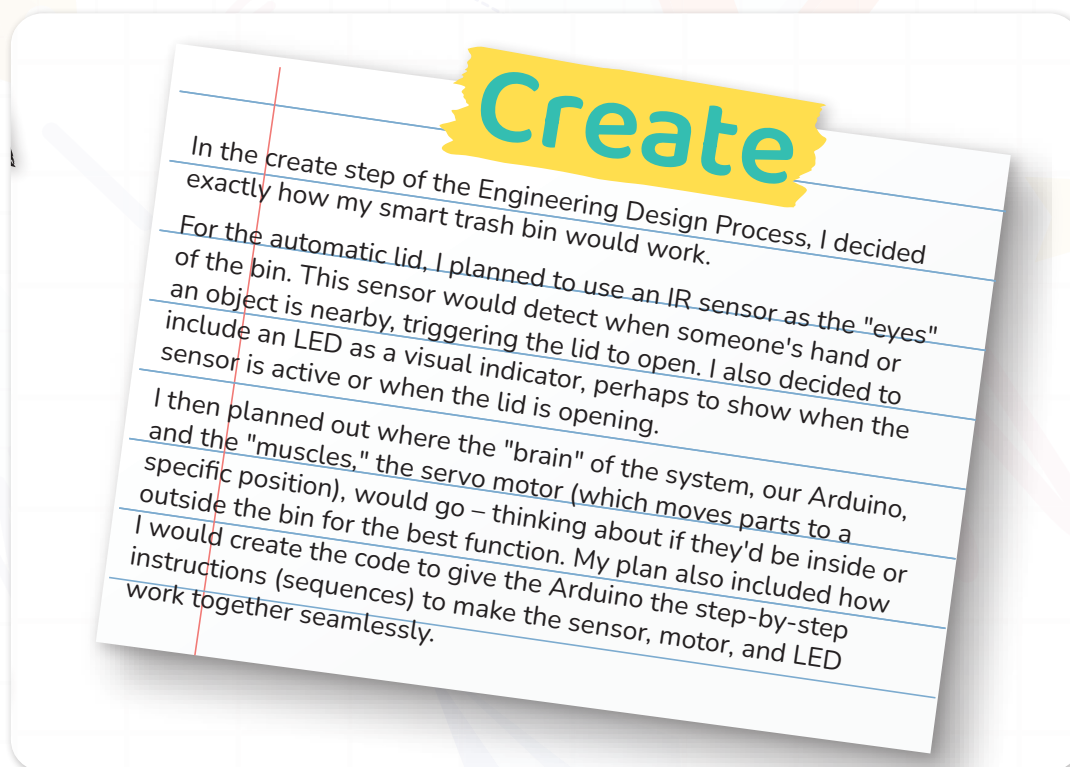

Activity Instructions:

1. Introduce the Puzzle Challenge: Begin by explaining the activity's relevance to creating automatic features, like the smart bin lid.
2. Explain Goal & Outcome: Clearly state the Goal: "Arrange the scrambled code lines into the correct order" and the Outcome: "When correctly sequenced, the circuit will turn on the LED and spin the servo when the IR sensor detects an object."
3. Distribute Code Snippets: Provide students with the scrambled code snippets(if they don't have the book.)
4. Guide Arrangement (Optional Initial Support): If students are new to coding, briefly review basic programming concepts like void setup() (for initial setup) and void loop() (for continuous execution). Discuss what variables might be (pinIR, pinServo, pinLED, val) and what they typically store.
5. Student Work: Allow students to work individually or in small groups to arrange the code. Encourage them to articulate their reasoning and discuss different approaches.
6. Review and Explain: Once students have attempted the puzzle, facilitate a class review of the correct sequence. Explain the purpose of each line of code, how the sensor reading affects the if/else logic, and how that logic controls the LED and servo motor.

Showcase Rubric:

Criteria	Excellent (4 Point)	Good (3 Points)	Satisfactory (2 Points)	Needs Improvement (1 Point)
Circuit Assembly	Correct wiring, no errors, all components functional.	Minor wiring issues, mostly functional.	Some errors, needs debugging.	Major wiring issues, not functional.
Code Implementation	Code is correct, optimized, and well-commented.	Code works but needs slight optimization.	Code runs with minor errors.	Code has multiple errors, does not run.
Lid Control	Lid opens/closes properly.	Minor response delays.	Works partially	Lid does not respond properly.
Presentation & Explanation	Clearly explains each component and logic.	Explains most parts, but lacks clarity in some areas.	Basic explanation with missing details.	Struggles to explain functionality.
Innovation & Problem Solving	Adds creative features beyond requirements.	Implements basic requirements successfully.	Minimal innovation, meets only basic needs.	Struggles to implement basic logic.

It's time to move to the Create part and add this.



Now I Can

objective:

Students will reflect on their learning by identifying the skills they used while designing and building a motorized merry-go-round, including creating a simple circuit, understanding motion and electricity, solving design problems, applying inclusive design principles, and using data to assess performance. This activity promotes metacognition, encourages ownership of learning, and reinforces STEM integration through engineering, science, and data interpretation.

Activity Instructions:

1. Read each statement aloud one by one and give students a moment to reflect silently.
2. Ask students to describe or act out what they built or coded that connects to each skill before checking the box.
3. Encourage students to leave a box unchecked if they're unsure—they can return to it after practicing more or asking for help.
4. Invite students to draw a mini diagram or write a keyword next to each checked skill (e.g., "IR sensor + servo," "lid opens," or a picture of their project).
5. Allow time for students to pair up or form small groups to discuss what each skill meant to them and how they applied it in their own project

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. (Students are designing a functional and accessible piece of equipment.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs. (Implicitly, students are designing a solution to the problem of inaccessible merry-go-rounds.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly. (Students present and explain their merry-go-round designs.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings. (The design plan serves as a visual display.)

3. Common Core State Standards (CCSS) - Mathematics (Geometry):

- **1.G.A.3:** Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) to create a composite shape, and compose new shapes from the composite shapes.